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In [1]: # KNN Student Result Prediction using Excel Data

# Step 1: Import Libraries
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
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In [2]: # Step 2: Read Excel file
data = pd.read_excel("STUDENT_DATA.xlsx")
```

```
In [3]: # Step 3: Display dataset
print("\nDataset:\n")
print(data)
```

Dataset:

	StudyHours	Attendance	Result
0	2	40	Fail
1	4	45	Fail
2	5	60	Pass
3	6	65	Pass
4	7	70	Pass
5	3	50	Fail

```
In [4]: # Step 4: Separate features and Label
X = data[['StudyHours', 'Attendance']] # Input features
y = data['Result'] # Target output
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In [5]: # Step 5: Create KNN Model (K = 3, Euclidean distance)
knn = KNeighborsClassifier(n_neighbors=3, metric='euclidean')
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In [6]: # Step 6: Train the model
knn.fit(X, y)
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Out[6]: KNeighborsClassifier
KNeighborsClassifier(metric='euclidean', n_neighbors=3)
```

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In [7]: # Step 7: New student data (Test data)
new_student = [[5, 55]] # StudyHours = 5, Attendance = 55
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In [8]: # Step 8: Predict result
prediction = knn.predict(new_student)
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C:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning: X does not have valid feature names, but KNeighborsClassifier was fitted with feature names
warnings.warn(
```

In [9]:

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# Step 9: Display result
print("\nNew Student Data (StudyHours=5, Attendance=55)")
print("Predicted Result:", prediction[0])
```

New Student Data (StudyHours=5, Attendance=55)
Predicted Result: Pass

In [10]: `import matplotlib.pyplot as plt`

In [11]:

```
# Separate Pass and Fail students
pass_students = data[data['Result'] == 'Pass']
fail_students = data[data['Result'] == 'Fail']

# Plot Pass students (Green)
plt.scatter(pass_students['StudyHours'],
            pass_students['Attendance'],
            color='green',
            label='Pass',
            s=100)

# Plot Fail students (Red)
plt.scatter(fail_students['StudyHours'],
            fail_students['Attendance'],
            color='red',
            label='Fail',
            s=100)

# Plot New Student (Blue Star)
plt.scatter(5, 55,
            color='blue',
            marker='*',
            s=250,
            label='New Student')
```

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# Labels and Title
plt.xlabel("Study Hours")
plt.ylabel("Attendance (%)")
plt.title("KNN Classification of Student Result")

# Legend
plt.legend()

# Show graph
plt.show()
```

